# MPG No. 36.4.147 $\bar{3}'m'$ (-3'1m' setting) [ Type III, trigonal ] [M tensor]

* Rank 0 tensor.

$$[M]$$

$$M = \mathbb{M}_{0}^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & M_{xx} & 0 \\ 0 & 0 & M_{zz} \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_{0}^{(1)} - \mathbb{M}_{u}^{(1)}$$
$$M_{zz} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{u}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & M_{xxzx} & 0 \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & 0 & -M_{xxzx} & 0 \\ M_{xxzz} & M_{xxzz} & M_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & -M_{xxzx} \\ M_{xxzx} & -M_{xxzx} & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & -M_{xxzx} & 0 & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_{u}^{(1)} - 4\mathbb{M}_{u}^{(2)}$$
$$M_{xxyy} = \mathbb{M}_{0}^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_{u}^{(1)}$$
$$M_{xxzz} = \mathbb{M}_{0}^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(1)}$$
$$M_{xxzx} = \mathbb{M}_{4c}^{(1)}$$
$$M_{zzzz} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_{u}^{(1)} + 8\mathbb{M}_{u}^{(2)}$$
$$M_{yzyz} = \mathbb{M}_{0}^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & M_{xxzz} & 0 & M_{xxzx} & 0 \\ 0 & 0 & M_{xxzz} & 0 & -M_{xxzx} & 0 \\ -M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzz} = 3\mathbb{M}_{u}^{(3)}$$
$$M_{xxzx} = -2\mathbb{T}_{3s}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)}$$
$$M_{xyxy} = \mathbb{M}_{0}^{(3)} + 4\mathbb{M}_{u}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & 0 \\ 0 & -M_{xxzx} & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & -M_{yzyz} & 0 \\ M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{T}_{3s}^{(2)}$$
$$M_{yzyz} = -3\mathbb{M}_{u}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = -3\mathbb{M}_{u}^{(6)}$$
$$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & M_{xxzx} & 0 & 0 & M_{xxxz} & 0 \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & 0 & -M_{xxzx} & 0 & 0 & -M_{xxxz} & 0 \\ M_{zzxx} & M_{zzxx} & M_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} & M_{yzzy} & 0 & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & 0 & M_{yzzy} & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & -M_{xxxz} & 0 & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} & -M_{xxzx} & 0 & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_{u}^{(1)} - 4\mathbb{M}_{u}^{(2)}$$
$$M_{xxyy} = \mathbb{M}_{0}^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_{u}^{(1)}$$
$$M_{xxzz} = \mathbb{M}_{0}^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(1)} + 3\mathbb{M}_{u}^{(3)}$$
$$M_{xxzx} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} + \mathbb{T}_{3s}^{(2)}$$
$$M_{xxxz} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} - \mathbb{T}_{3s}^{(2)}$$
$$M_{zzxx} = \mathbb{M}_{0}^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(1)} - 3\mathbb{M}_{u}^{(3)}$$
$$M_{zzzz} = \mathbb{M}_{0}^{(1)} + 2\mathbb{M}_{0}^{(2)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_{u}^{(1)} + 8\mathbb{M}_{u}^{(2)}$$
$$M_{yzyz} = \mathbb{M}_{0}^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(2)} - 3\mathbb{M}_{u}^{(5)}$$
$$M_{yzxy} = -\mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)}$$
$$M_{yzzy} = \mathbb{M}_{0}^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_{u}^{(2)} + 3\mathbb{M}_{u}^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} & M_{yzzy} & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & -M_{yzzy} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)} - 3\mathbb{M}_{u}^{(6)}$$
$$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$$
$$M_{yzzy} = -\mathbb{M}_{0}^{(3)} + 2\mathbb{M}_{u}^{(4)} - 3\mathbb{M}_{u}^{(6)}$$
$$M_{xyxy} = \mathbb{M}_{0}^{(3)} + 4\mathbb{M}_{u}^{(4)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxxz} \\ 0 & M_{xxxx} & 0 \\ 0 & 0 & M_{zzzz} \\ -M_{xxxz} & 0 & M_{yyzz} \\ M_{zzxx} & 0 & 0 \\ 0 & M_{xxyy} & 0 \\ 0 & M_{yzzy} & 0 \\ M_{xxxz} & 0 & -\frac{M_{xxxx}}{2} + \frac{3M_{xxyy}}{2} + M_{yyzz} - M_{yzzy} + M_{zzxx} \\ \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} & 0 & -M_{xxxz} \\ 0 & -M_{xxxz} & 0 \end{bmatrix}$$

$$M_{xxxx} = 3\mathbb{M}_0^{(1)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$
$$M_{xxxz} = \mathbb{M}_{4c}^{(1)}$$
$$M_{zzzz} = 3\mathbb{M}_0^{(1)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{yyzz} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)}$$
$$M_{zzxx} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)}$$
$$M_{yzzy} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)}$$